

ME23312 ENGINEERING THERMODYNAMICS

Unit II – Second Law and Availability Analysis · Presentation 1 of 10

Why a Second Law? Heat Engine, Refrigerator and Heat Pump

Limitations of the first law · thermal energy reservoirs · cyclic devices · thermal efficiency
· coefficient of performance · $(COP)_{HP} = (COP)_{ref} + 1$

In this presentation

1. What Unit II covers
2. Processes have a direction – what the first law misses
3. What the second law gives us
4. Thermal energy reservoirs – source and sink
5. Cyclic devices

1. Heat engine and thermal efficiency
2. Refrigerator and its COP
3. Heat pump and its COP
4. Relation between the two COPs

What Unit II covers

- Heat engine, refrigerator, heat pump; the two statements of the second law
- Carnot cycle, Carnot theorem, Clausius inequality
- Entropy, T–s diagram, Tds equations; principle of increase of entropy
- High- and low-grade energy; availability, irreversibility, second-law efficiency

KEY POINT Course outcome CO2: analyse the second law – work, heat, entropy and availability – for engineering applications.

1

Why a second law is needed

Processes have a direction

- Hot coffee always cools to room temperature – it never warms itself from the cooler room
- A falling weight turning a paddle heats water – heating the water never lifts the weight
- A gas expands to fill an evacuated vessel – it never collects itself back into one half
- None of the reverse processes would violate the first law – yet they never happen

Two limitations of the first law

No direction

- Both directions of a process satisfy energy conservation
- Yet only one direction is ever observed
- Natural processes are not reversible

No quality of energy

- One kJ of work and one kJ of heat are treated as equal
- Work converts completely into heat; heat never converts completely and continuously into work
- Heat at high temperature is worth more than the same heat at low temperature

What the second law gives us

- The **direction** in which processes proceed
- An **upper limit** to the fraction of heat that can be converted into work
- An **absolute temperature scale** independent of any substance
- The property **entropy** – the measure of irreversibility
- **Availability and irreversibility** – the quality of energy and the loss of work potential

2

Thermal energy reservoirs

Thermal energy reservoir

- A body of very large heat capacity that absorbs or supplies unlimited heat **without any appreciable change in temperature**
- Atmosphere, oceans, rivers and lakes; a controlled furnace; a large block of ice
- Any body whose heat capacity is very large compared with the heat exchanged
- Exchanges heat reversibly: its entropy change is simply Q/T

Source and sink

Source – supplies heat

- Furnace of a boiler
- Combustion gases in an engine
- The sun

Sink – receives heat

- Condenser cooling water
- The atmosphere
- A river

Source or sink – a matter of function

The same atmosphere is a sink for a power plant and a source for a heat pump.

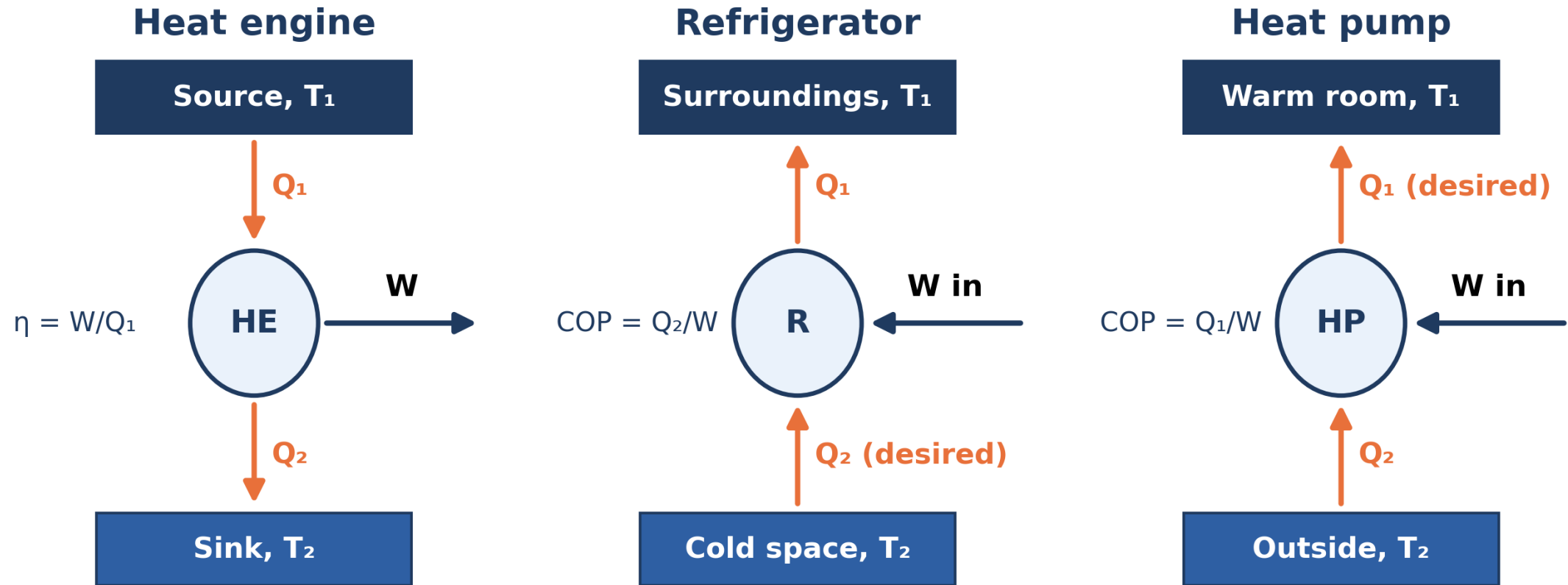
- The distinction is by what the reservoir does in the process, not by its nature

3

Heat engine, refrigerator and heat pump

The three cyclic devices

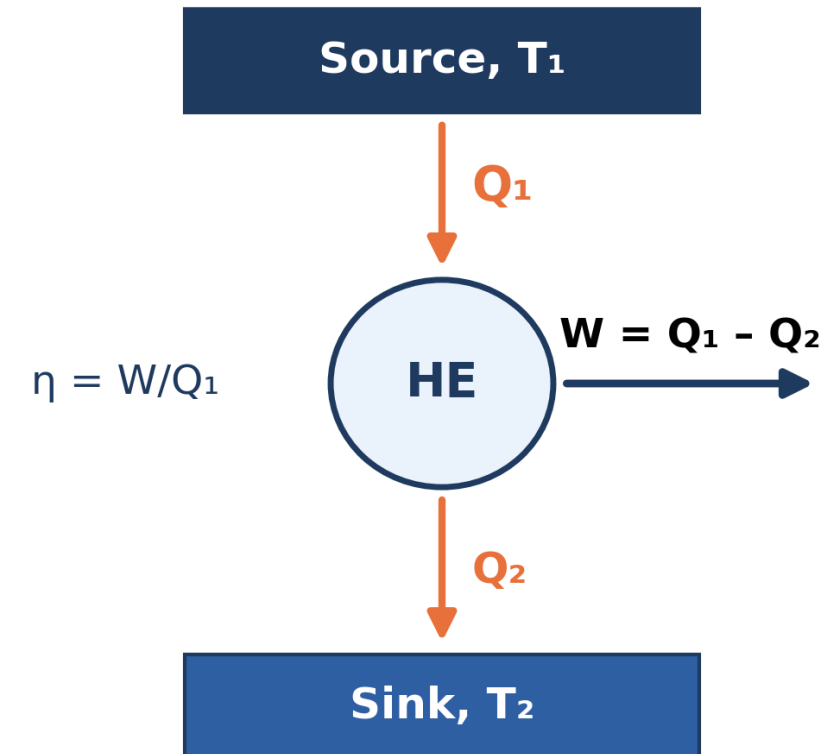
The three cyclic devices



The working fluid executes a cycle repeatedly – its state at the end of each cycle equals that at the beginning

Heat engine

- Receives heat Q_1 from a source at T_1 , converts part into net work W , rejects the rest Q_2 to a sink at T_2
- Working fluid returns to its initial state each cycle: $\Delta U = 0$, so $\mathbf{W_{net} = Q_1 - Q_2}$
- Steam plant: boiler receives Q_1 , turbine gives W_T , condenser rejects Q_2 , pump takes W_P



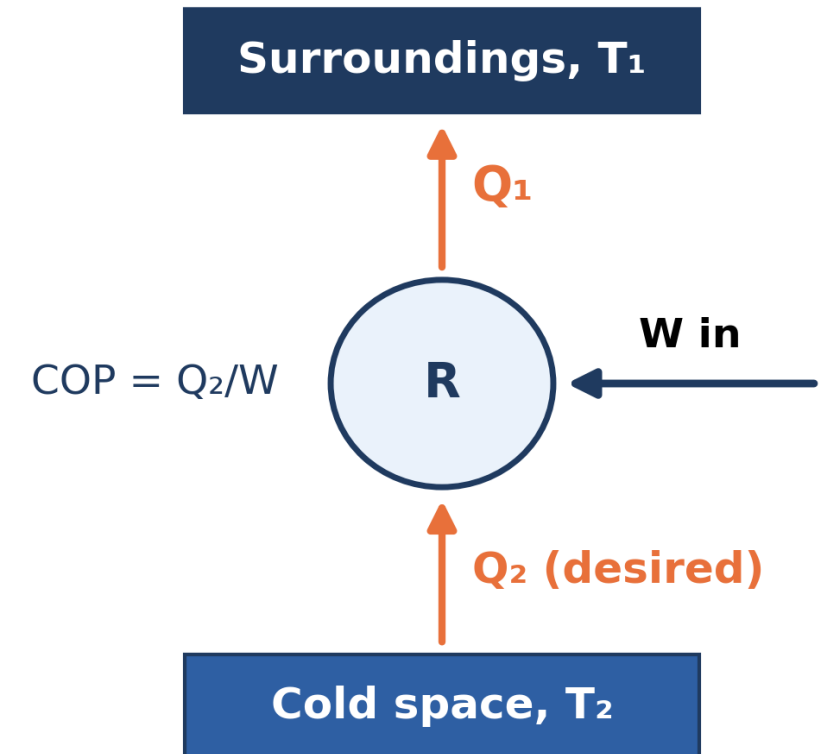
Thermal efficiency of a heat engine

$$\eta_{th} = W_{net}/Q_1 = (Q_1 - Q_2)/Q_1 = 1 - Q_2/Q_1$$

- Desired output (net work) ÷ required input (heat supplied)
- Q_2 is never zero, so the efficiency is **always less than unity**
- Typical values: 0.25 to 0.45

Refrigerator

- Removes heat Q_2 from a cold space at T_2 and rejects Q_1 to the surroundings at T_1 , using a work input W
- First law for the cycle: $Q_1 = Q_2 + W$
- **Desired effect:** the heat removed from the cold space, Q_2
- Working fluid: the refrigerant



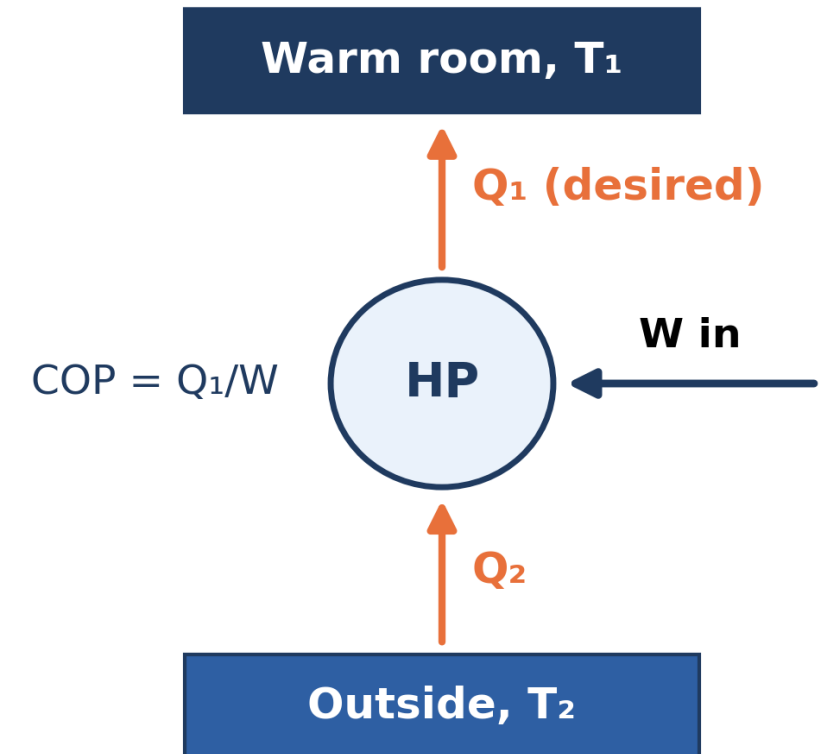
Coefficient of performance of a refrigerator

$$\begin{aligned} (\text{COP})_{\text{ref}} &= \text{desired effect} / \text{work input} \\ &= Q_2 / W = Q_2 / (Q_1 - Q_2) \end{aligned}$$

- Not called an efficiency because it is normally **greater than unity** – typically 2 to 4
- **Tonne of refrigeration:** heat removal that freezes 1000 kg of water at 0 °C to ice in 24 h; 1 TR $\approx$ 3.87 kW (3.5 kW is also used)

Heat pump

- The same cyclic device as a refrigerator, but with a different purpose
- Delivers heat Q_1 to a warm space (a room in winter) by absorbing Q_2 from the cold outside air or ground, with work input W
- **Desired effect:** the heat delivered to the warm space, Q_1



Coefficient of performance of a heat pump

$$(\text{COP})_{\text{HP}} = Q_1/W = Q_1/(Q_1 - Q_2)$$

- Typical values: 3 to 5
- A heat pump with COP 4 delivers 4 kJ of heating for every kJ of electricity – 3 kJ are "pumped" free from the outside air

Relation between the two COPs

$$\begin{aligned} (\text{COP})_{\text{HP}} &= Q_1/W = (Q_2 + W)/W = Q_2/W + 1 \\ (\text{COP})_{\text{HP}} &= (\text{COP})_{\text{ref}} + 1 \end{aligned}$$

- The COP of a heat pump is therefore **always greater than unity**
- Even with $Q_2 = 0$ the work itself is delivered as heat: $\text{COP} = 1$ – exactly what a resistance heater achieves

The three devices compared

Device	Purpose	Desired effect	Performance	Typical
Heat engine	Produce work from heat	$W_{\text{net}} = Q_1 - Q_2$	$\eta_{\text{th}} = 1 - Q_2/Q_1$	0.25 – 0.45
Refrigerator	Keep a space cold	Q_2 removed from cold space	$(\text{COP})_{\text{ref}} = Q_2/(Q_1 - Q_2)$	2 – 4
Heat pump	Keep a space warm	Q_1 delivered to warm space	$(\text{COP})_{\text{HP}} = Q_1/(Q_1 - Q_2)$	3 – 5

Refrigerator and heat pump – the same machine

A refrigerator and a heat pump are physically the same machine – compressor, condenser, expansion valve and evaporator. They differ only in what we want from them.

- **Refrigerator:** valued for the heat it **removes** at low temperature;
Heat pump: for the heat it **delivers** at high temperature
- That is why the two COPs differ by exactly one

Summary

- First law: no direction, no quality of energy – the second law supplies both
- Reservoir: huge heat capacity, constant temperature; source supplies, sink receives
- Heat engine: $W = Q_1 - Q_2$; $\eta_{th} = 1 - Q_2/Q_1 < 1$
- Refrigerator: $(COP)_{ref} = Q_2/W$; 1
 $TR \approx 3.87 \text{ kW}$
- Heat pump: $(COP)_{HP} = Q_1/W = (COP)_{ref} + 1 > 1$
- Same machine, different desired effect

Next: Statements of the second law

- Kelvin–Planck and Clausius statements; the PMM2
- Equivalence of the two statements
- Reversibility revisited – causes of irreversibility